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TREND-BASED MOCK PAPER

GATE Civil Engineering

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Civil Engineering
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Volume of sphere r:

- (A) $\frac{4}{3} < \pi r^3$
- (B) $< \pi r^3$
- (C) $< \pi r^2$
- (D) $2 <$

Q2. [1-MCQ] [1 mark]

Choose similar to 'diligent':

- (A) Lazy
- (B) Industrious
- (C) Careless
- (D) Ignorant

Section B: Civil Engineering

Q3. [1-MCQ] [1 mark]

Order of $dy/dx=y$:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q4. [1-NAT] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q5. [2-NAT] [2 marks]

The vector (1,2,3) projected onto (2,1,0) is:

(Numerical Answer Type — Enter the numerical value)

Q6. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
 - (B) $y = C/x$
 - (C) $y = C \cdot x^2$
 - (D) $x^2 + y^2 = C$
-

Q7. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q8. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
 - (B) $y = x \cdot \ln(x)$
 - (C) $y = x \cdot \sin(\ln x)$
 - (D) $y = x^2$
-

Q9. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)} \cdot u(t)$ is:

- (A) $1/(s+3)$
 - (B) $1/(s-3)$
 - (C) $1/s$
 - (D) $e^{-3/s}$
-

Q10. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
 - (B) 0
 - (C) -1
 - (D) $\cos(\pi)$
-

Q11. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q13. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) $\pi \cdot r^2$
- (B) $2 \cdot \pi \cdot r$
- (C) $\pi \cdot r$
- (D) $2 \cdot \pi \cdot r^2$

Q14. [2-MCQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q15. [2-NAT] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q17. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q18. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q19. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q20. [2-NAT] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q21. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q22. [2-MSQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q23. [2-NAT] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q25. [2-MSQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q26. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q27. [2-MCQ] [2 marks]

$P(|Z| < 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q28. [1-MCQ] [1 mark]

Strongest acid:

- (A) HCl
- (B) HNO₃
- (C) H₂SO₄
- (D) HClO₄

Q29. [1-MCQ] [1 mark]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1

(D) -1

Q30. [2-MCQ] [2 marks]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q31. [2-MCQ] [2 marks]

First-order half-life with $k=0.693 \text{ min}^{-1}$:

- (A) 1 min
- (B) 0.693 min
- (C) 1.44 min
- (D) 0.5 min

Q32. [2-NAT] [2 marks]

pH of 10^{-8} M HCl approx:

(Numerical Answer Type — Enter the numerical value)

Q33. [1-NAT] [1 mark]

Cantilever UDL w over L. Deflection:

(Numerical Answer Type — Enter the numerical value)

Q34. [1-NAT] [1 mark]

Max shear stress in solid shaft:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-MCQ] [2 marks]

Fixed-free Euler buckling:

- (A) $\pi^2 EI / L^2$
- (B) $\pi^2 EI / (4L^2)$
- (C) $4\pi^2 EI / L^2$
- (D) $2\pi^2 EI / L^2$

Q36. [2-MSQ] [2 marks]

SS beam UDL w N/m, span L. Max BM:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL^2/4$
- (D) $wL^2/2$

Q37. [2-MCQ] [2 marks]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q38. [2-NAT] [2 marks]

Camber on road provides:

(Numerical Answer Type — Enter the numerical value)

Q39. [2-MCQ] [2 marks]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q40. [1-MCQ] [1 mark]

Influence line for SS beam reaction:

- (A) Triangle
- (B) Rectangle
- (C) Linear decreasing
- (D) Parabola

Q41. [2-MCQ] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

- (A) 0.57
- (B) 0.43
- (C) 0.5
- (D) 0.75

Q42. [2-MSQ] [2 marks]

Cantilever UDL w , L . Fixed end moment:

- (A) $wL^2/2$
- (B) wL^2
- (C) $wL^2/4$
- (D) $wL^2/8$

Q43. [1-MCQ] [1 mark]

Darcy-Weisbach for:

- (A) Minor losses
- (B) Major losses
- (C) Both
- (D) None

Q44. [1-MCQ] [1 mark]

Rainfall measured by:

- (A) Hygrometer
- (B) Rain gauge
- (C) Anemometer
- (D) Barometer

Q45. [1-NAT] [1 mark]

Manning's equation for:

(Numerical Answer Type — Enter the numerical value)

Q46. [1-MCQ] [1 mark]

Gumbel distribution used for:

- (A) Rainfall
- (B) Flood frequency
- (C) Runoff
- (D) None

Q47. [1-NAT] [1 mark]

Compound interest: P at r% for 2 years =

(Numerical Answer Type — Enter the numerical value)

Q48. [1-MCQ] [1 mark]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q49. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q50. [1-MCQ] [1 mark]

LL=60%, PL=30%. PI =

- (A) 30
- (B) 60
- (C) 90
- (D) 30%

Q51. [1-NAT] [1 mark]

Terzaghi q_{ult} for $c=0$, $\phi=30^\circ$, $B=2m$, $D_f=1m$, $\gamma=18$:

(Numerical Answer Type — Enter the numerical value)

Q52. [2-MCQ] [2 marks]

Rankine K_a for $\phi=30^\circ$:

- (A) $1/3$
- (B) 0.33
- (C) 0.43
- (D) 0.75

Q53. [1-MCQ] [1 mark]

Moment-area method gives:

- (A) Slope
- (B) Deflection
- (C) Both
- (D) None

Q54. [1-MCQ] [1 mark]

Bending moment in UDL simply supported:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL/2$
- (D) wL^2

Q55. [1-MCQ] [1 mark]

Method of joints gives:

- (A) Forces only
- (B) Moments
- (C) Both
- (D) None

Q56. [1-MCQ] [1 mark]

Bearing capacity of footing:

- (A) Depends on soil
- (B) Depends on load
- (C) Both
- (D) None

Q57. [1-NAT] [1 mark]

Atterberg limits:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-MCQ] [1 mark]

BOD stands for:

- (A) Biological Oxygen Demand
- (B) Basic Oxygen Demand
- (C) Bacterial Oxygen Demand
- (D) None

Q59. [1-NAT] [1 mark]

Sedimentation removes:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

PCU stands for:

- (A) A
- (B) B
- (C) C
- (D) D

Q61. [1-NAT] [1 mark]

Camber provided for:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

Dumpy level used for:

- (A) Angle

- (B) Distance
- (C) Levelling
- (D) All

Q63. [1-NAT] [1 mark]

Theodolite measures:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-NAT] [1 mark]

Initial setting time OPC:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-NAT] [1 mark]

Water-cement ratio affects:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	A	1
2	1MCQ	B	1
3	1MCQ	B	1
4	1MCQ	C	1
5	1MCQ	A	1
6	1MCQ	A	1
7	1MCQ	C	1
8	1NAT	D	1
9	1MCQ	B	1
10	1MCQ	B	1
11	1NAT	B	1
12	1MCQ	B	1
13	1MCQ	A	1

14	1NAT	B	1
15	1MCQ	A	1
16	1NAT	B	1
17	1MCQ	C	1
18	1NAT	B	1
19	1NAT	A	1
20	1NAT	C	1
21	1NAT	A	1
22	1MCQ	D	1
23	1MCQ	A	1
24	1MCQ	C	1
25	1NAT	A	1
26	1NAT	A	1
27	1NAT	B	1
28	1MCQ	A	1

29	1NAT	A	1
30	1MCQ	B	1
31	2NAT	B	2
32	2MCQ	B	2
33	2NAT	A	2
34	2MCQ	A	2
35	2MCQ	A	2
36	2MCQ	B	2
37	2NAT	B	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2MCQ	A	2
41	2MCQ	B	2
42	2MCQ	B	2
43	2MCQ	C	2

44	2NAT	B	2
45	2NAT	A	2
46	2MSQ	A	2
47	2MSQ	A	2
48	2MCQ	A	2
49	2MCQ	A	2
50	2MCQ	A	2
51	2NAT	C	2
52	2MCQ	A	2
53	2MCQ	B	2
54	2MCQ	A	2
55	2MCQ	C	2
56	2MCQ	A	2
57	2NAT	A	2
58	2NAT	D	2

59	2MCQ	A	2
60	2MSQ	C	2
61	2NAT	B	2
62	2MCQ	B	2
63	2MSQ	A	2
64	2MCQ	A	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Differential Equations | Difficulty: difficult

Q4.

Answer: C

Solution:

Correct Answer: C

Subject: Structural Engineering | Topic: Deflection | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Structural Engineering | Topic: Beams | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Structural Engineering | Topic: Trusses | Difficulty: moderate

Q7.

Answer: C

Solution:

Correct Answer: C

Subject: Geotechnical Engineering | Topic: Foundation | Difficulty: moderate

Q8.

Answer: D

Solution:

Correct Answer: D

Subject: Geotechnical Engineering | Topic: Soil Mechanics | Difficulty: difficult

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Pipe Flow | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Precipitation | Difficulty: easy

Q11.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Open Channel | Difficulty: easy

Q12.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Flood | Difficulty: moderate

Q13.

Answer: A

Solution:

Correct Answer: A

Subject: Environmental Engineering | Topic: Sewage | Difficulty: moderate

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Environmental Engineering | Topic: Water | Difficulty: moderate

Q15.

Answer: A

Solution:

Correct Answer: A

Subject: Transportation Engineering | Topic: Traffic | Difficulty: easy

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Transportation Engineering | Topic: Highways | Difficulty: moderate

Q17.

Answer: C

Solution:

Correct Answer: C

Subject: Surveying & Geomatics | Topic: Levelling | Difficulty: moderate

Q18.

Answer: B

Solution:

Correct Answer: B

Subject: Surveying & Geomatics | Topic: Theodolite | Difficulty: moderate

Q19.

Answer: A

Solution:

Correct Answer: A

Subject: Construction Materials | Topic: Cement | Difficulty: difficult

Q20.

Answer: C

Solution:

Correct Answer: C

Subject: Construction Materials | Topic: Mix Design | Difficulty: moderate

Q21.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q22.

Answer: D

Solution:

Perchloric acid (HClO_4) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q23.

Answer: A

Solution:

$$PI = LL - PL = 60 - 30 = 30.$$

Subject: Geotechnical | Topic: Geotechnical | Difficulty: easy

Q24.

Answer: C

Solution:

$$R_A = (L-x)/L. \text{ Linear decreasing from } L \text{ to } 0.$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q25.

Answer: A

Solution:

$$q_{ult} = q_{Nq} + 0.5 \cdot \gamma \cdot B \cdot N_{\gamma} = 18 \cdot 22.5 + 0.5 \cdot 18 \cdot 2 \cdot 19.7 = 405 + 354.6 = 759.6 \text{ kPa}.$$

Subject: Geotechnical | Topic: Geotechnical | Difficulty: moderate

Q26.

Answer: A

Solution:

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|. \text{ Equality when } u \text{ and } v \text{ are linearly dependent.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q27.

Answer: B

Solution:

$$\text{Cantilever UDL: } \delta = wL^4/(8EI).$$

Subject: SOM | Topic: SOM | Difficulty: moderate

Q28.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q29.

Answer: A

Solution:

$$\tau_{max} = T \cdot c/J = 16T/(\pi \cdot d^3).$$

Subject: SOM | Topic: SOM | Difficulty: easy

Q30.

Answer: B

Solution:

$$\delta_n = 2 - 4 = -2.$$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q31.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q32.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q33.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000 \text{ MPa}$.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q35.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x \cdot v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: B

Solution:

$K=2$, $L_{eff}=2L$. $P_{cr} = \pi^2 EI / (2L)^2 = \pi^2 EI / (4L^2)$.

Subject: SOM | Topic: SOM | Difficulty: moderate

Q37.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: difficult

Q38.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

$DF_{AB} = 4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q40.

Answer: A

Solution:

$K_a = \tan^2(45-15) = \tan^2(30) = 1/3$.

Subject: Geotechnical | Topic: Geotechnical | Difficulty: moderate

Q41.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q42.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q43.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q45.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: A

Solution:

$FEM = wL \cdot (L/2) = wL^2/2$ (hogging).

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q47.

Answer: A

Solution:

SS UDL: max BM at center = $wL^2/8$.

Subject: SOM | Topic: SOM | Difficulty: easy

Q48.

Answer: A

Solution:

$t_{1/2} = 0.693/k = 0.693/0.693 = 1$ min.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q49.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q50.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x \cdot \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q52.

Answer: A

Solution:

Row 2 = $2 \cdot$ Row 1, Row 3 = $3 \cdot$ Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q53.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q54.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q55.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q57.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q58.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{\text{total}}$ approx 1.1×10^{-7} . pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q59.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50} [1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q60.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q63.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q64.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q65.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 25 questions, 48 marks
Chemistry: 5 questions, 8 marks
SOM: 4 questions, 6 marks
Civil Engineering: 3 questions, 6 marks
Structural Analysis: 3 questions, 5 marks
Water Resources Engineering: 4 questions, 4 marks
Miscellaneous: 3 questions, 4 marks
Geotechnical: 3 questions, 4 marks
Structural Engineering: 3 questions, 3 marks
General Aptitude: 2 questions, 2 marks
Geotechnical Engineering: 2 questions, 2 marks
Environmental Engineering: 2 questions, 2 marks
Transportation Engineering: 2 questions, 2 marks
Surveying & Geomatics: 2 questions, 2 marks
Construction Materials: 2 questions, 2 marks

Difficulty Distribution:

Easy: 19 | Moderate: 36 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.